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EDUCATION

Cornell University, College of Arts & Sciences *August 2023–May 2027*
Degrees: Bachelor of Arts in Computer Science, Bachelor of Arts in Astronomy *Ithaca, NY*
GPA: 4.023 / 4.3
Relevant Coursework: Computational Imaging, Computer Graphics, Graduate Computer Graphics & Vision Seminar, Machine Learning, Compilers, Multiwavelength Astronomical Techniques, Planetary Surface Processes, Physics of the Planets, Modeling and Machine Learning in Astronomy, Systems Programming, Quantum Computing, Galaxies & Cosmology, Engineering Mathematics Sequence, Introductory Physics Sequence, Analysis of Algorithms, Functional Programming, Discrete Structures, Object-Oriented Programming, Interactive Computer Graphics (Spring 2027)

RELEVANT EXPERIENCE

Cornell Computational Imaging Lab *January 2026–Present*
Undergraduate Research Assistant under Kristina Monakhova *Ithaca, NY*

- Applying differentiable rendering techniques to refractive optical systems, allowing calibration of lens misalignments.
- **Summer 2026:** Employed as a full-time student researcher in the Cornell Bowers Undergraduate Research Experience (BURE) program for ten weeks; presented my work at a concluding program-wide symposium.
- Utilizing Python's ecosystem of numerical and astronomical packages, including PyTorch, for efficient computation.
- Conducting thorough literature reviews of papers in signal processing, optics, physically-based rendering, and astronomy.
- Meeting with collaborators across fields, including Lisa Kaltenegger of the Cornell astronomy department.

IEEE International Conference on Computational Photography (ICCP) 2026 *July 2026*
Summer School Fellow and Main Conference Attendee *Princeton, NJ*

- Was selected as one of only twenty-five fully-funded "fellows" for the second annual ICCP Summer School.
- Attended two days of collaborative, hands-on courses focused on modern imaging hardware and software.
- Constructed a working Michelson interferometer and an adaptive optics setup, and used Cryo-EM software and SPADs.
- Attended the presentations, poster sessions, and meals of the main conference following the Summer School.

Cornell ALPS (Algorithms for Large-Scale Parallel Systems) Lab *February–December 2025*
Undergraduate Research Assistant under Giulia Guidi *Ithaca, NY*

- Accelerated genomic analysis algorithms using sparse linear algebra and concurrency techniques in heterogeneous memory.
- **Summer 2025:** Employed full-time in the BURE program.
- Developed in CUDA C++, collected data using Bash and Python, and utilized external libraries for CPU/GPU acceleration.
- Utilized software such as Slurm on NERSC's Perlmutter supercomputer to develop, execute, and benchmark computations.

CS 3410 (Computer Systems Organization & Programming) *August–December 2025*
Undergraduate Teaching Assistant *Ithaca, NY*

- Reworked previously problematic assignments, developed assignment handout materials, and graded student submissions.
- Co-led weekly in-person Lab sections where students were led through difficult concepts from lecture.
- Held weekly in-person office hours to assist students with conceptual understanding and assignment completion.

ASTRO 6580 (Planetary Surface Processes Field Trip) *March 30–April 3, 2026*
Field Trip Participant *Death Valley and Mojave Desert, CA & NV*

- Applied planetary science concepts in understanding the formation processes of myriad geologic features in the area.
- Used GIS software to make predictions about landforms before the trip, and then tested these predictions in the field.
- Kept a comprehensive field notebook detailing local geologic formations and their theorized origins.
- Presented relevant concepts to the class in the field, and authored a synthesizing report at the conclusion of the trip.

PERSONAL & COURSE PROJECTS

CS 4120/4121 (Compilers) Project *January–May 2026*

Compiler for the ρ (Rho) and η (Eta) programming languages *Ithaca, NY*

- Engineered an optimizing compiler for η, a statically-typed imperative language, and later the object-oriented extension ρ.
- Communicated clearly and frequently with two teammates about goals, work division, and implementation.
- Carefully constructed Java code for all stages of compilation, including code optimization, in a containerized environment.
- Co-authored several comprehensive engineering reports detailing technical specifications and issues encountered.

ASTRO 4434 (Physics of the Planets) Final Project *October–December 2026*

Jovian Radio Emission Simulation as an Exoplanet Analogue *Ithaca, NY*

- Utilizing the existing ExPRES library written in IDL to simulate CMI emissions from Jupiter's magnetic field, with a partner.
- Consulting the literature to form accurate hypotheses about how radio emissions can be used to infer field properties.
- Co-authoring a 10–15-page final manuscript detailing our hypotheses, methods, and findings.

CS 4620 (Computer Graphics) Final Project *December 2025*

Tim, Destroyer of Worlds *Ithaca, NY*

- Developed an interactive 3D "planet destruction" demo in less than a week, coordinating within a small group of students.
- Extended a course-provided TypeScript library for 3D rendering, itself based on Three.js, to develop our demo.
- Implemented and utilized graphics techniques including marching cubes, procedural generation, and signed distance fields.

ASTRO 6577 (Planetary Surface Processes) Final Project *May 2026*

Enceladus Ice Jet Particle Simulator *Ithaca, NY*

- Worked with a partner to create a 3D particle simulation of the icy jets in the South Polar Terrain of Saturn's moon Enceladus.
- Consulted existing literature to derive physically-accurate particle emission geometry and parameters for ~100 individual jets.
- Engineered an efficient, multithreaded particle simulation program in MATLAB using geometric matrix operations.
- Presented in-class progress updates, drawing conclusions from 3D visualizations of relevant quantities for ejected particles.

CS 4621 (Computer Graphics Practicum) Project *August–December 2025*

"4Dungeon" *Ithaca, NY*

- Created a video game with four spatial dimensions in the Unity game engine, collaborating in a self-led team of five students.
- Developed the underlying infrastructure and set of primitives for four-dimensional rendering in a three-dimensional engine.
- Presented project milestone progress to course instructors and peers throughout the semester.

CS 3110 (Functional Programming) Final Project *October–December 2024*

"PSBop" (link to GitHub repository) *Ithaca, NY*

- Engineered a study-tracking web app in OCaml in a team of three CS 3110 students under self-direction.
- Utilized SQLite bindings in OCaml to create a persistent database system in the back-end.
- Communicated clearly and frequently with teammates in-person and via Slack about project goals and development.
- Wrote and edited reports detailing collaborative progress and our technical obstacles and advances.

Video Game Project with the Cornell Development in Games Association *July–August 2024*

"Completely Hammered" (link to itch.io page) *Remote*

- Developed a full video game over the course of five weeks, coordinating with a team of around twelve Cornell students.
- Became closely familiar with GitHub and Unity as I assembled game objects and programmed game components in C#.
- Created and released a substantial update for the game with two other team members following the initial release.

Personal Todo List Manager Project *May 2026–Present*

"todo" (link to GitHub repository) *Various Locations*

- Independently authoring an interactive command-line "todo list" task management system using the Rust language.
- Progressively adding useful features to a simple baseline, refactoring using modular software engineering techniques.
- Using Rust's powerful type system to create highly general functions and data structures, enabling large-scale code reuse.

TECHNICAL SKILLS

Programming Languages: C/C++, Rust, Java, Bash, OCaml, Python, Julia, C#, TypeScript, RISC-V Assembly, MATLAB, Nushell
Mathematics: Multivariable calculus, differential equations, linear algebra, signal processing, optimization, statistics, probability
Physics: Mechanics, special relativity, electromagnetism, waves and oscillations, quantum physics, orbital mechanics
Development & Creative Software: Git, Slurm, GNU/Linux, Docker/Podman, Visual Studio Code, MATLAB, Jupyter, Unity, Godot, QGIS, Helix, LaTeX, Nix, NixOS, uv, Pixi, Anaconda, Cargo, Obsidian, Aseprite, Krita, Inkscape, p5.js, FreeCAD
Software Libraries: Numpy, PyTorch, Scipy, Astropy, Matplotlib, DeepLens, CUDA, cuSPARSE, OpenMP, MPI, Three.js, Plots.jl
Observatory Telescopes: Hartung–Boothroyd Observatory optical telescope, Cornell Space Sciences Building 3.8-m radio telescope
Spoken Languages: English (native), español (low intermediate), Gayogo̲hǫ̀ǫ́néha:ʔ (novice)

EXTRACURRICULARS & LEADERSHIP

Cornell University Big Red Marching and Pep Bands *August 2023–Present*
Member (trumpet, mellophone, and trombone), Webmaster, Historian, Librarian, Spinmaster *Ithaca, NY*

- Attending rehearsals and exhibitions including parades and musical performances before, during, and after sporting events.
- Took photos at events, managed the websites of both bands, and created software to streamline music printing.
- Serving as one of the leaders of the trumpet section and fostering an inclusive and welcoming environment for all members.

Cornell University Wind Symphony *August 2025–May 2026*
Member (trumpet) *Ithaca, NY*

- Attended two rehearsals each week, in addition to dress rehearsals, in preparation for two public concerts per semester.
- Participated in sectionals, in which I and fellow trumpet players worked through difficult excerpts.
- Practiced the repertoire outside of rehearsal time, taking private lessons within Cornell's department of music.